

Documentation file for Do-files and Datasets

Replication files for “Seasonal liquidity, rural labor markets and agricultural production”

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Data sources:

1. Survey data: the data for the analysis comes from 7 surveys conducted from September 2013-September 2015. See Table 1 and Appendix C of the paper for further detail.
 - a. Baseline (November/December 2013), Harvest year 1 (July-September 2014) and Endline year 2 (July-September 2015) surveys at the household level containing most of the non-labor variables
 - b. Labor and midline surveys (January 2014-August 2015) at the household level that captured household labor allocation (four rounds)
 - c. A midline maize assessment (February-March 2014) that included maize measurement and visual recording (photographs)
 - d. Maize price surveys (September 2013-May 2015) that captured maize price data from vendors in Chipata.
2. Administrative Data (not included):
 - a. Zambia DHS data from the 2001/2, 2007 and 2013/14 survey rounds.
 - i. Data on the fraction of children under 5 considered underweight.
 - ii. Accessed through IPUMS DHS in October 2018. Redistribution is prohibited, so data are not provided but can be obtained from <http://idhsdata.org>. Full citations are provided below and in the text.
 - iii. Used to create Appendix figure B.1.
 - b. Data from the Msakera Agricultural Research Station in Chipata District, Zambia:
 - i. Monthly rainfall data from 1970 through 2015, by growing season
 - ii. Data obtained in-person in January 2016 from the Ministry of Agriculture’s research station in Chipata District. The address of the station is Farm D25 off Great East Road, Chipata 10101, Zambia. Full citations are provided below and in the text.
 - iii. Data are used to create Appendix figure B.4.

Description of the data:

The data files are broken into two folders: “Clean subject panels” and “Analysis.” The datasets in the former were created by cleaning and de-identifying the data from the surveys above and

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grouping the variables into smaller, subject-themed panels (e.g., the labor panel contains all of the variables capturing labor data from the baseline, midline, harvest, labor, and endline surveys). All datasets except for “price_survey.dta” (which is measured for vendors) contain variables measured at the household level. “hhid” is the unique identifier for households, variables with “mem_id” identify members of the household, and “vid” identifies the village. The datasets with “admin” in their name contain information on which surveys the households participated in and whether they were assigned to a treatment (as well as which one).

The data files in “Analysis” are the ones used to create the figures and graphs in “Seasonal liquidity, rural labor markets and agricultural production” and its appendix. The folder is comprised of the final version of the admin datasets, a re-combined panel of the household-level variables, and the final version of the price survey data (all trimmed to only include variables pertinent to the analysis).

A full list of the datasets included is located at the end of this document. All variables are labeled in Stata. Any questions regarding the exact wording of survey questions can be directed to the authors.

Reuse and Citation:

Please cite this dataset if you are re-using it.

Description of the Do-Files:

The “Do Files” folder consists of eight scripts. “Final_datasets.do” takes the clean subject panels as inputs and outputs the analysis files. “Model_simulation_figures.do” includes code to simulate the theoretical model included in “Seasonal liquidity, rural labor markets and agricultural production.” The other do-files produce the figures or tables indicated by their names (e.g., “Main_tables.do” reproduces the tables in the main paper). Appendix_figure_B1.do and Appendix_figure_B4.do require DHS and rainfall data, respectively, that are not included in the archive, so are posted as separate files that include both the necessary cleaning of the data and the code for producing the figures.

General organization of top folder:

1. Data
 - a. Analysis
 - b. Clean subject panels
2. Do Files
3. Master.do
4. Output
 - a. Appendix Figures
 - b. Appendix Tables
 - c. Main Figures
 - d. Main Tables

e. Simulation Figures

Note: you will need to create the “Output” folder along with its subsidiary folders

Replication Instructions:

In order to replicate the analyses found in “Seasonal liquidity, rural labor markets and agricultural production,” first create the folders missing from the top file structure laid out above. Then run the master do-file found in the top file structure. The master file calls each of the six files in “Do Files” in the correct order and outputs the tables and figures to their respective sub-folder.

There are two non-default Stata packages used in the analysis:

1. leebounds
2. ineqdec0 (note the zero at the end; ineqdeco is a similar but different package)

The last replication was run in Stata 16.1.

References:

Central Statistical Office, Central Board of Health [Zambia] and ORC Macro. Zambia Demographic and Health Survey 2001-02 [Dataset]. Data Extract from ZMKR42.SAV. IPUMS Demographic and Health Surveys (IPUMS DHS), version 7, IPUMS and ICF [Distributors]. Accessed from <http://idhsdata.org>

Central Statistical Office, Ministry of Health, Tropical Diseases Research Centre, University of Zambia [Zambia] and ORC Macro. Zambia Demographic and Health Survey 2007 [Dataset]. Data Extract from ZMKR51.SAV. IPUMS Demographic and Health Surveys (IPUMS DHS), version 7, IPUMS and ICF [Distributors]. Accessed from <http://idhsdata.org>

Central Statistical Office, Ministry of Health [Zambia], and ICF International. Zambia Demographic and Health Survey 2013-14 [Dataset]. Data Extract from ZMKR61.SAV IPUMS Demographic and Health Surveys (IPUMS DHS), version 7, IPUMS and ICF [Distributors]. Accessed from <http://idhsdata.org>

Msekera Research Station. 1970–2015. “Monthly Rainfall Data.” Zambia Ministry of Agriculture [dataset].

Full List of Datasets:

Analysis:

1. admin_outcomes_final.dta: Analysis dataset containing data on outcomes, survey inclusion, and treatment assignment
2. household_panel_final.dta: Dataset containing most of the variables used in analysis; at the household level and includes variables from the clean subject panels

3. price_survey_final.dta: Contains vendor price data used in analysis
4. reporting_bias_final.dta: Contains data on reporting bias used in analysis
5. yield_checks_final.dta: Contains data on maize yield checks used in analysis

Clean Subject Panels:

1. Admin_clean.dta: Small version of “admin” datasets
2. Admin_outcomes_clean.dta: Fuller “admin” dataset; used to produce “admin_outcomes_final.dta”
3. Cashflow_panel_full_clean.dta: HH level variables on cashflow; contains both long and short recall versions
4. Cashflow_panel_long_recall_clean.dta: The long recall variables from the above dataset
5. Cashflow_panel_short_recall_clean.dta: The short recall variables from the above dataset
6. Consumption_full_panel_clean.dta: HH level variables on consumption
7. Employer_survey_clean.dta: HH level variables on employers of HH members
8. Health_panel_clean.dta: HH level variables on member health
9. Hh_controls.dta HH level variables used as controls in analysis
10. Labor_panel_full_clean.dta: HH level variables on labor; includes both long and short recall variables
11. Labor_panel_long_recall_clean.dta: long recall version of the above dataset
12. Labor_panel_short_recall_clean.dta: short recall version of the above dataset
13. Price_survey: Data on maize prices in Chipata
14. Production_function_clean.dta: HH level data on agricultural production
15. Productivity_panel_clean.dta: HH level data on agricultural productivity
16. Reporting_bias_clean.dta: HH level data on social desirability index
17. SavingBorrowing_long_recall_clean.dta: HH level data on savings, long recall version
18. SavingBorrowing_full_panel_clean.dta: HH level data on savings; contains both long and short recall variables
19. SavingBorrowing_short_recall_clean.dta: HH level data on savings, short recall version
20. Wage_panel_clean.dta: HH level data on wages
21. Yield_checks_clean.dta: complete dataset of verification of maize yields